

Schizophrenia Classification from EEG Signals: A Comparative Study of Classical Learning, Deep Networks and Time-Series Foundation Models

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Abstract—Schizophrenia lacks consolidated objective biomarkers, and electroencephalography (EEG) offers a non-invasive avenue for its detection. This work presents a rigorous comparative study on the public *ASZED* dataset (153 subjects: 76 patients and 77 controls) that confronts three model families: (i) classical machine learning over tabular features extracted with `tsfresh`; (ii) deep networks over time–frequency representations (a custom CNN, ResNet18, EfficientNet and a Vision Transformer); and (iii) Amazon’s *Chronos* time-series foundation model used as a frozen encoder. The entire protocol enforces a strict subject-wise partition to eliminate the information leakage that is endemic in biomedical time series. Exploratory analysis reveals a marked *cortical disconnection* (functional connectivity dropping from >0.8 in controls to $0.3\text{--}0.7$ in patients) and a *hyperfrontality* pattern that motivates an optimal six-channel fronto-temporal montage. The best overall model is Chronos, which—with the full montage and subject-level aggregation—attains an *F1-Score* of 0.846, *Accuracy* of 0.826 and *ROC-AUC* of 0.826, together with the strongest window-level discrimination (*ROC-AUC* = 0.859). Strikingly, a foundation model never trained on EEG surpasses both bespoke deep networks and exhaustively optimized classical pipelines, while the latter remain remarkably competitive at a fraction of the computational cost. We release the full reproducible pipeline as open source.

Index Terms—EEG, schizophrenia, time series, foundation models, Chronos, Vision Transformer, feature engineering, SHAP, subject-wise partition, time–frequency analysis.

I. INTRODUCTION

SCHIZOPHRENIA is a chronic psychiatric disorder whose diagnosis remains fundamentally clinical and therefore subject to inter-rater variability. The search for objective biomarkers has positioned electroencephalography (EEG) as an attractive modality: it is non-invasive, inexpensive, and captures the cortical electrical dynamics—where the connectivity and rhythmicity alterations hypothesized for the disorder are expressed—at high temporal resolution.

The information-leakage trap. Machine learning applied to EEG is plagued by a recurrent and severe methodological bias: *data leakage*. When the temporal windows of a single subject are scattered at random across the training and test sets, the model can memorize the idiosyncratic *fingerprint* of

the individual rather than learn the clinical marker, which dramatically inflates the reported performance. The only methodologically valid scheme is the *subject-wise partition*: all windows of a subject belong to exactly one split. This study adopts that partition strictly and reports, in addition to window-level metrics, metrics aggregated at the subject level, which constitute the clinical unit of interest.

From bespoke models to foundation models. Two paradigms dominate EEG decoding: hand-crafted feature engineering feeding classical learners, and end-to-end deep networks operating on time–frequency images. A third, emerging paradigm—*time-series foundation models* pre-trained on massive heterogeneous corpora—has not, to our knowledge, been systematically evaluated on clinical EEG. We close this gap by benchmarking Chronos in a zero-/few-shot regime against both established paradigms.

Contributions. This paper makes five contributions: (1) a reproducible, leakage-free protocol on ASZED; (2) an exploratory analysis that quantifies the cortical-disconnection and hyperfrontality hypotheses and grounds the channel-selection decision; (3) a comprehensive comparison of eleven architectures across three paradigms, including the first zero-/few-shot evaluation of the Chronos foundation model on clinical EEG; (4) an ablation over the time–frequency representation (STFT vs. CWT vs. FFT) and the number of channels (six vs. eighteen); and (5) a clinical interpretation of the selected features via SHAP values.

II. MATERIALS AND METHODS

A. Dataset and subject-wise partition

ASZED comprises EEG recordings from 153 subjects. The recordings come from two acquisition formats that share the 19 electrodes of the international 10–20 system; we harmonized them by channel name so that no subject is discarded for format reasons. After preprocessing, 5,946 valid windows were obtained (11 recordings were discarded for insufficient length). Table I summarizes the stratified 70/15/15 subject-wise partition; class proportions are preserved across splits, and no subject appears in more than one split.

Table I
ASZED SUMMARY AND STRICT SUBJECT-WISE PARTITION.

	Train	Val	Test	Total
Subjects	107	23	23	153
Windows (5 s)	4,333	749	864	5,946
Patient ratio (class 1)	0.529	0.439	0.436	—

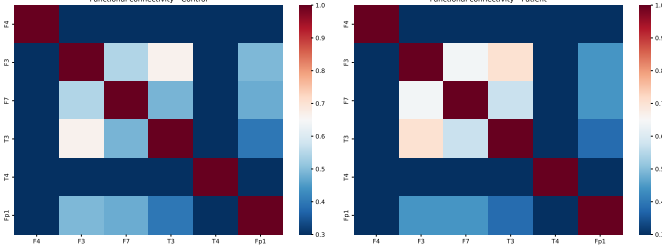


Figure 1. Functional connectivity (average inter-channel correlation). The control brain (left) is highly synchronous (> 0.8); the patient brain (right) exhibits the characteristic drop to 0.3–0.7 (cortical disconnection).

B. Preprocessing

Each recording underwent a standard clinical pipeline. A band-pass filter of 0.5–45 Hz removes slow electrode drift (high-pass edge) and high-frequency electromuscular interference (low-pass edge); a 50 Hz *notch* filter suppresses mains noise; signals are resampled to a common 250 Hz; they are segmented into fixed 5-second windows (1,250 samples); and finally each channel is normalized via *z-score*. Each window is thus a $19 \times 1,250$ tensor. The choice of a 45 Hz low-pass edge keeps the five clinical bands (delta, theta, alpha, beta and low gamma) while staying well below the 125 Hz Nyquist limit imposed by the 250 Hz sampling rate.

C. Exploratory analysis and spatial reduction

The exploratory data analysis (EDA) surfaced two clinically meaningful phenomena that guided the design.

Cortical-disconnection hypothesis. The functional-connectivity matrices (average inter-channel Pearson correlation, Fig. 1) show a *healthy* brain that is highly synchronous, with dominant correlations above 0.8. In patients the synchrony degrades markedly toward the 0.3–0.7 range, consistent with the functional disconnection hypothesis of the disorder. This loss of long-range coherence is one of the most reproducible electrophysiological findings in schizophrenia and directly motivates connectivity-aware feature design.

Hyperfrontality phenomenon. The Welch power spectral density (PSD) and a Welch’s *t*-test applied to the per-channel PSD variance identified the fronto-temporal regions as the most discriminative. On this basis we discarded the global 18/19-channel montages in favor of an optimal six-channel *Top-6* montage {F4, F3, F7, T3, T4, Fp1}. This decision is also economical: it reduces the input dimensionality of the feature branch by $\sim 68\%$. The PCA+UMAP projection (Fig. 2) and spectral clustering over these six channels reveal a partial but non-trivial class separability, anticipating a moderately hard task under the strict subject-wise partition.

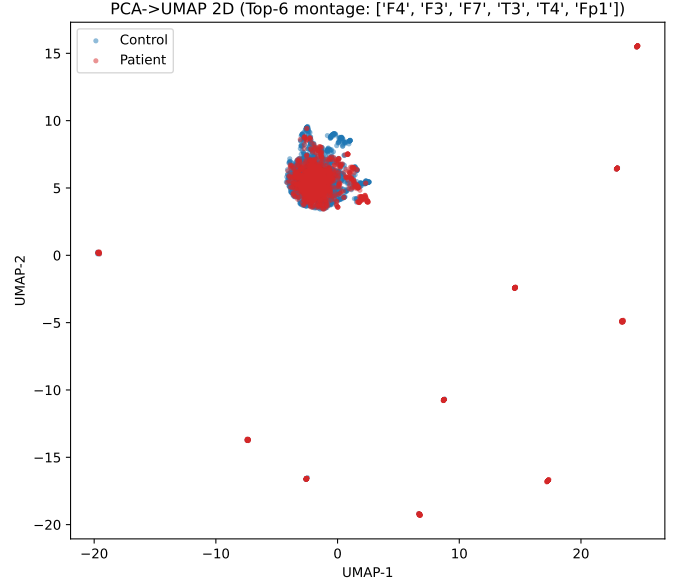


Figure 2. PCA+UMAP 2D projection of the windows (Top-6 montage). The partial separability between controls and patients anticipates the difficulty of classification under a strict subject-wise partition.

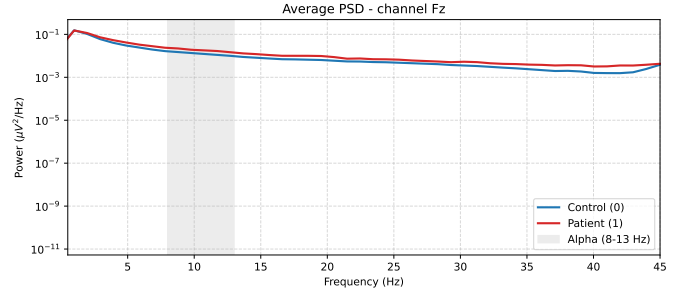


Figure 3. Average power spectral density (Welch) at Fz, controls vs. patients (logarithmic scale). The shaded band marks the alpha rhythm (8–13 Hz).

Spectral signatures. Three complementary spectral views refine the picture. The average power spectral density (PSD) at Fz (Fig. 3) contrasts the two classes across the 0.5–45 Hz band on a logarithmic scale, exposing where the spectral power diverges and highlighting the alpha rhythm. The temporal grand average (Fig. 4) shows the mean *z*-scored waveform per class at representative fronto-central channels, with a shaded dispersion band; the near-zero means with class-dependent dispersion are themselves informative, as they reflect the non-stationary, phase-incoherent nature of resting EEG—no fixed temporal template exists, which is precisely why marginal and higher-order descriptors, rather than the raw waveform, carry the signal. Finally, the relative band-power boxplots (Fig. 5) quantify the proportion of energy in each clinical rhythm. Empirically, patients exhibit a *rightward spectral shift*: a reduced share of slow delta activity ($\Delta_{\text{rel}} = -0.066$) compensated by increased beta ($+0.018$) and, most notably, gamma ($+0.031$) power. This relative high-frequency excess is consistent with the gamma-band dysregulation reported in schizophrenia and complements the loss of long-range synchrony seen in the connectivity matrices.

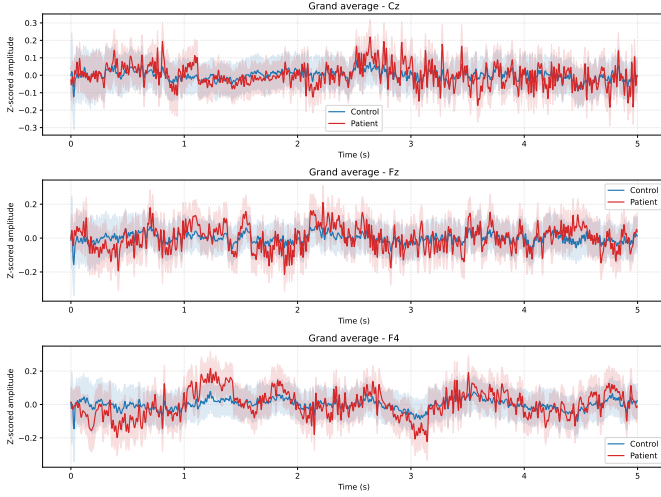


Figure 4. Temporal grand average (z -scored) at fronto-central channels with 0.1σ dispersion bands. The class-dependent dispersion—rather than the mean—reflects the non-stationary character of resting EEG.

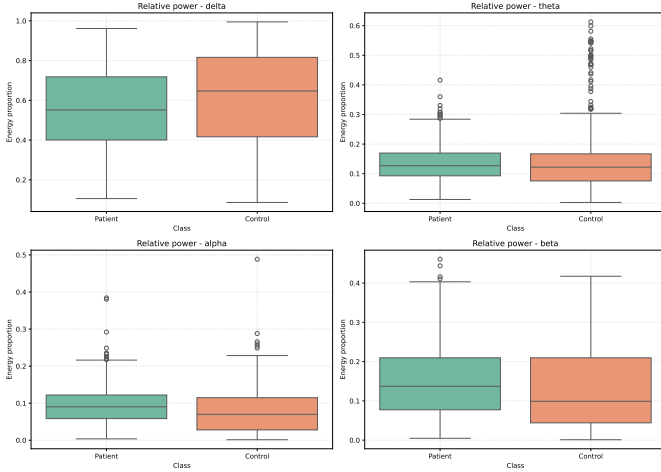


Figure 5. Relative band-power distributions per class. Patients show a reduced delta share and increased beta/gamma share—a rightward spectral shift.

D. Time–frequency transform: justification of the STFT

For the deep branch, windows were converted into time–frequency images. We prioritized the Short-Time Fourier Transform (STFT, spectrograms) over the Continuous Wavelet Transform (CWT, scalograms) for two reasons. First, *computational*: generating CWT scalograms over the full window set entails tens of thousands of independent multi-scale transforms (on the order of $5,946$ windows $\times$ channels $\times$ 40 scales, i.e. more than a million scale–channel convolutions), which is impractical on local hardware; we therefore restricted the CWT to the Top-6 montage and treated it as an ablation branch. Second, *empirical*: as documented in Section V, the CWT did *not* outperform the STFT. The STFT imposes a fixed time–frequency resolution (the Heisenberg–Gabor limit), which for the 0.5–45 Hz band and 5-second windows proved an adequate compromise for this task. Spectrograms are log-compressed and z -scored using statistics fit on the training set only.

Table II
FEATURE-SELECTION FUNNEL (TOP-6 MONTAGE).

Stage	# Features
tsfresh (EfficientFCParameters)	4,662
Statistical relevance (FDR)	1,800
Variance filter	1,800
Collinearity filter ($ r > 0.85$)	1,204
SHAP-driven selection (≥ 0.1)	51

III. TABULAR FEATURE ENGINEERING AND ADVANCED SELECTION

The classical branch rests on a principle central to the time-series course: the raw signal is a poor representation for standard learners. A window $\mathbf{X} \in \mathbb{R}^N$ with $N = 1,250$ samples per channel lives in a high-dimensional space where distance metrics degrade (the curse of dimensionality), and its samples are strongly autocorrelated, violating the *independent and identically distributed* (IID) assumption of most algorithms. Feature extraction resolves both problems by mapping $\mathbf{X}$ to a compact, interpretable descriptor vector $\phi(\mathbf{X}) \in \mathbb{R}^d$.

A. Memory-efficient iterative extraction

Over the six Top-6 channels we extracted features with `tsfresh` under the `EfficientFCParameters` dictionary, which yields 4,662 features per window. The naive extraction of all channels at once exhausts memory (*OOM*). We implemented an *iterative channel-by-channel* strategy: each channel is processed in isolation, its feature sub-table is persisted to disk in `.parquet` format (which doubles as a resumable cache), and memory is explicitly released with `gc.collect()` before the next channel. The sub-tables are concatenated only at the end. The per-channel cache, keyed by electrode name, further allowed us to reuse the six already computed channels when expanding the study to eighteen.

B. Selection funnel

The massive table was reduced through a rigorous filtering funnel, fit *exclusively* on the training set (Table II). A statistical relevance filter (`tsfresh` FDR test) discards features with no univariate association with the label; a variance filter removes near-constant columns; a collinearity filter drops one of every pair with $|r| > 0.85$; and a model-driven step retains the features whose normalized aggregate SHAP magnitude—computed by combining the importances of a LightGBM and a Random Forest—reaches ≥ 0.1 . The funnel compresses $4,662 \rightarrow 51$ final features, with no additional PCA required.

IV. ARCHITECTURES AND EVALUATED MODELS

Classical machine learning. Over the 51 selected features we trained five models: Logistic Regression, SVM, Random Forest, XGBoost [8] and LightGBM [9]. Features were standardized (fit on training only) for the scale-sensitive models.

Deep learning. The deep networks consume the STFT spectrograms, treating the EEG channels as image channels. We evaluated a *custom CNN* trained from scratch; a Vision

Table III

SUBJECT-LEVEL RESULTS, TOP-6 MONTAGE ($n=23$ TEST SUBJECTS), SORTED BY F1-SCORE. ACC: ACCURACY; PREC: PRECISION; SENS: SENSITIVITY; SPEC: SPECIFICITY; BACC: BALANCED ACCURACY.

Model	Acc	Prec	Sens	Spec	F1	AUC	BAcc
ViT (STFT)	0.783	0.688	1.000	0.583	0.815	0.803	0.792
ResNet18	0.739	0.667	0.909	0.583	0.769	0.826	0.746
EfficientNet	0.696	0.643	0.818	0.583	0.720	0.735	0.701
EEG-Conformer	0.652	0.588	0.909	0.417	0.714	0.727	0.663
Random Forest	0.652	0.600	0.818	0.500	0.692	0.674	0.659
LightGBM	0.652	0.600	0.818	0.500	0.692	0.705	0.659
XGBoost	0.652	0.600	0.818	0.500	0.692	0.727	0.659
Logistic Reg.	0.652	0.600	0.818	0.500	0.692	0.705	0.659
SVM	0.652	0.615	0.727	0.583	0.667	0.727	0.655
CNN (custom)	0.565	0.533	0.727	0.417	0.615	0.652	0.572

Transformer (ViT) [5] and an EfficientNet-B0 [6] via `timm`; and a ResNet18 [4] with transfer learning. For the latter, the first convolution (`conv1`) was adapted to a six-channel input by initializing the new weights as the average of the three pre-trained RGB filters; the backbone was frozen except for the last convolutional block (`layer4.requires_grad=True`) and the classifier head—a deep fine-tuning strategy that adapts the model to the EEG domain without overfitting. We additionally included the EEG-Conformer [11] (convolution+attention) operating directly on the raw signal.

Chronos as a universal encoder. We integrated Amazon’s *Chronos* time-series foundation model [7] (`chronos-t5-small`) as a *frozen* encoder: each window was embedded channel by channel and pooled (mean over time and channels) into a 512-dimensional vector, on top of which a linear Logistic Regression head was trained. This configuration evaluates the zero-/few-shot capability of a model that has never seen EEG.

V. RESULTS

A. Global comparison (Top-6 montage)

Table III reports the subject-level performance—the clinical unit—over the 23 test subjects. The ViT leads by F1-Score (0.815) with perfect sensitivity, followed by ResNet18 (0.769, ROC-AUC 0.826). At the window level (not tabulated for brevity), Chronos attains the highest ROC-AUC (0.859) and Accuracy (0.784), and the ViT the highest F1 (0.755). Subject aggregation systematically raises the metrics (e.g. the ViT rises from F1 0.755 to 0.815) by averaging out the window-level noise within each patient, acting as an implicit ensemble.

B. Channel ablation: Top-6 vs. Top-18

Table IV contrasts the two montages at subject level (image models use the STFT representation). The picture is nuanced: most models improve *modestly* when moving from six to eighteen channels—the custom CNN and EEG-Conformer benefit the most—yet the single best Top-6 model (ViT, F1 0.815) is *not* surpassed by its own Top-18 STFT variant. Crucially, the window-level ROC-AUC of Chronos is essentially identical across montages (0.859 with Top-6 vs. 0.856 with eighteen channels). This quantitatively corroborates

Table IV

CHANNEL ABLATION (SUBJECT LEVEL): TOP-6 VS. TOP-18. IMAGE MODELS USE THE STFT REPRESENTATION. [†]CHRONOS TOP-6 SUBJECT-LEVEL METRICS WERE UNAVAILABLE; ITS WINDOW-LEVEL ROC-AUC WAS 0.859 (TOP-6) VS. 0.856 (TOP-18).

Model	Top-6		Top-18	
	F1	AUC	F1	AUC
ViT (STFT)	0.815	0.803	0.741	0.727
ResNet18	0.769	0.826	0.769	0.841
EfficientNet	0.720	0.735	0.769	0.780
CNN (custom)	0.615	0.652	0.750	0.841
EEG-Conformer	0.714	0.727	0.800	0.773
Random Forest	0.692	0.674	0.720	0.674
LightGBM	0.692	0.705	0.692	0.705
XGBoost	0.692	0.727	0.692	0.689
Logistic Reg.	0.692	0.705	0.692	0.765
SVM	0.667	0.727	0.720	0.735
Chronos [†]	—	—	0.846	0.826

Table V

REPRESENTATION ABLATION (TOP-18, SUBJECT LEVEL): SUBJECT-LEVEL F1-SCORE / ROC-AUC PER ARCHITECTURE AND TIME-FREQUENCY REPRESENTATION. BEST F1 PER ROW IN BOLD.

Architecture	STFT	CWT	FFT
CNN (custom)	0.750 /0.841	0.640/0.705	0.741/0.811
ResNet18	0.769/0.841	0.692/0.750	0.786 /0.788
ViT	0.741/0.727	0.786/0.735	0.815 /0.826
EfficientNet	0.769/0.780	0.714/0.674	0.786 /0.811

that the discriminative information is concentrated in the fronto-temporal region: six well-chosen channels suffice, and adding twelve more yields diminishing returns while tripling the feature-extraction cost.

C. Representation ablation: STFT vs. CWT vs. FFT

Table V isolates the effect of the time–frequency representation for the four image architectures (Top-18, subject level). The answer to our ablation question is clear: *wavelets do not win*. The STFT is the strongest representation at the window level, while the FFT is surprisingly strong at the subject level (e.g. ViT-FFT reaches F1 0.815); the CWT consistently ranks last within each architecture. We note a conservative caveat: the CWT operated on six channels versus eighteen for STFT/FFT, a bias *against* the wavelet representation that does not overturn the qualitative ordering.

D. Error structure: confusion matrices

The aggregate metrics hide *which* errors each model commits; the subject-level confusion matrices (Fig. 6) expose them. The test set comprises 23 subjects (12 controls, 11 patients). Three regularities stand out. First, the error structure is markedly *asymmetric*: false negatives are rare across the board (0–3 of 11 patients missed), so sensitivity is near-perfect, whereas the errors concentrate on *false positives* (controls flagged as patients), which is why specificity is the limiting metric. Second, the foundation model attains the best operating point: Chronos at eighteen channels yields *zero* false negatives and only four false positives (TN=8, FP=4, FN=0, TP=11), catching

every patient while misclassifying the fewest controls. Third, moving from six to eighteen channels primarily *reduces false alarms*: the EEG-Conformer and the custom CNN both raise their true negatives from 5 to 8, and the Random Forest from 6 to 7, with sensitivity essentially unchanged. The extra channels therefore help the models *reject* controls rather than *detect* patients, who were already being caught.

Clinically, this high-sensitivity / moderate-specificity profile is exactly the desirable operating point for a *screening triage* tool: a missed patient (false negative) is far costlier than a false alarm, which merely escalates a control to confirmatory review. The models err on the safe side, and the residual false positives concentrate—as the head-to-head analysis below confirms—on the genuinely borderline subjects.

E. Qualitative analysis

Fig. 7 shows the channel-averaged STFT spectrograms of four control and four patient subjects from the test set, each annotated with the ViT’s mean predicted probability of belonging to the patient class. Two qualitative patterns emerge that mirror the EDA. Control spectrograms tend to concentrate energy in a narrower, more structured low-frequency band with smoother temporal evolution, whereas several patient spectrograms display more diffuse, higher-entropy energy spread across the band—visually consistent with the loss of rhythmic organization. The ViT probabilities track these visual cues sensibly: controls receive low scores and patients high scores in the majority of cases, while the misclassifications correspond precisely to the visually ambiguous spectrograms, illustrating both the strengths and the failure modes of the image-based models.

F. Training dynamics

Fig. 8 shows the optimization curves of the STFT deep models and makes the overfitting argument visually explicit. The custom CNN and the EfficientNet drive the training loss down while the validation loss diverges after only a handful of epochs, triggering early stopping at the dotted line; the pre-trained ViT and ResNet18, by contrast, exhibit a far more stable validation trajectory and optimize for more epochs before plateauing. This is the optimization-level counterpart of Table III: deprived of transferred priors, a high-capacity network trained on only 107 subjects memorizes the training windows rather than the clinical marker, whereas transfer learning regularizes the trajectory and yields genuine generalization.

G. Model complementarity

The three paradigms are not merely ranked—they are partially *complementary*. Fig. 9 contrasts, per test subject, the probabilities of the signal-based EEG-Conformer and the tabular SVM. Although both reach comparable aggregate scores, their errors are partially disjoint: each model correctly classifies two subjects that the other misclassifies (the off-diagonal quadrants), while they agree on thirteen and jointly fail on six. The disagreements are not random; they concentrate on the borderline subjects whose probabilities sit near the 0.5

threshold. This partial error decorrelation between a deep time-domain model and a classical spectral–statistical model indicates that the two capture distinct, complementary facets of the pathology, and motivates a future stacked ensemble that could exceed any single model—a promising direction precisely because the foundation model, the vision Transformers and the tabular learners err on different subjects.

VI. DISCUSSION

A. Why a foundation model wins

The central finding is that Chronos—frozen and never trained on EEG—achieves the best overall performance. This is best understood through the lens of the time-series course. Chronos tokenizes a (scaled) numerical series into a finite vocabulary and processes it with a T5 encoder pre-trained on enormous, heterogeneous corpora, effectively learning a *language* of temporal dynamics: trends, autocorrelation structure, regime changes and local stationarity. These are exactly the inductive priors that hand-crafted descriptors approximate one at a time. Because EEG, despite its clinical specificity, is still a numerical time series, those transferred priors apply, and a simple linear head suffices to read them out. The result reframes EEG decoding as a *representation-transfer* problem rather than an architecture-search problem, and suggests that the marginal returns of training ever-larger bespoke networks on a few hundred subjects are limited.

B. Why classical pipelines remain competitive

The SHAP-guided classical models form a remarkably robust mid-tier. Random Forest, XGBoost, LightGBM and Logistic Regression *tie* at subject-level F1 (0.692) and, notably, *beat* the CNN trained from scratch (0.615). In other words, 51 interpretable features and a negligible computational budget match or exceed a multi-million-parameter convolutional network trained without pre-training. The decisive advantage of the deep tier comes solely from the *pre-trained* models (ViT, ResNet18) and the foundation model (Chronos). This dissociation—transfer of knowledge matters, raw capacity does not, under limited clinical data—is the practical lesson for the field: an interpretable, auditable classical baseline should always accompany deep results, both as a sanity check against leakage and as a deployable low-cost alternative.

C. The selected features as a course-level reading of the pathology

The 51 retained descriptors are a compact dictionary of the time-domain concepts studied in the course, and their identities are clinically eloquent (Fig. 11). The dominant features are the *variance* and *absolute energy* (F3, T3, F4), direct proxies of cortical power and the substrate of the hyperfrontality finding. Next come *permutation entropy* ($m=5$; F3, F7, F4, Fp1), a nonlinear-dynamics descriptor that quantifies the irregularity/complexity of the series and rises with the disorganized neural processing of the disorder. The funnel also retained the *first-order autoregressive coefficient* AR(1) (T4), a direct estimate of the autocorrelation structure ρ_k , and the

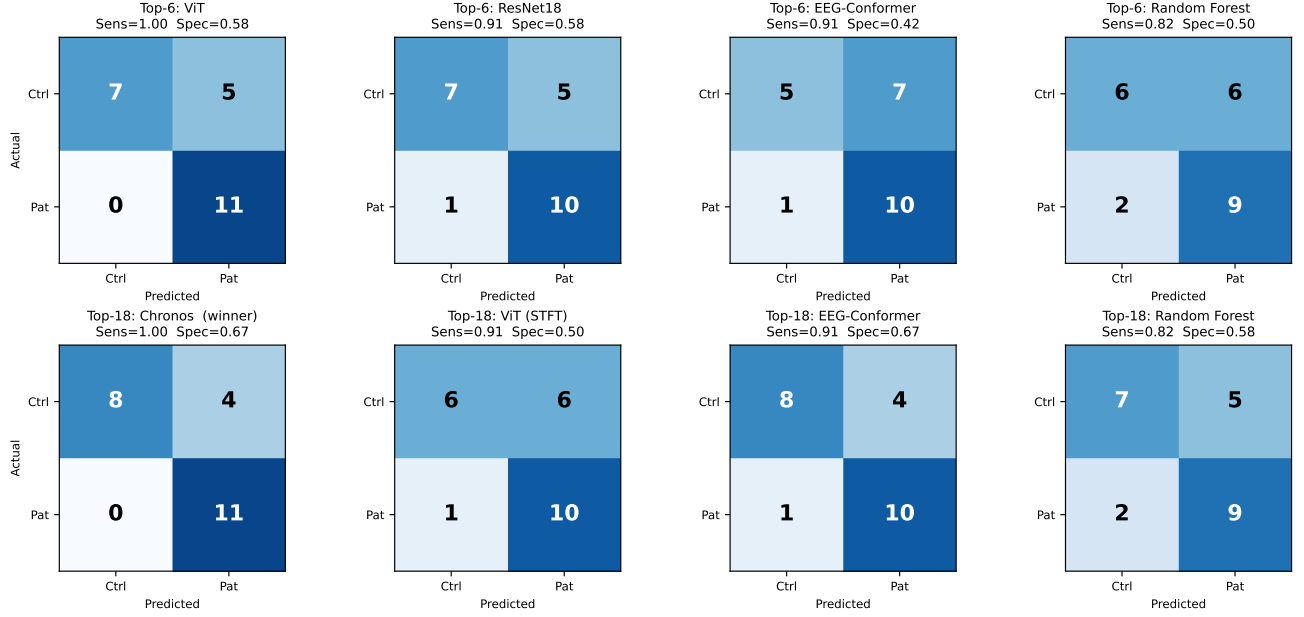
Subject-level confusion matrices (TEST, $n=23$: 12 controls, 11 patients): Top-6 (top) vs Top-18 (bottom)

Figure 6. Subject-level confusion matrices on the TEST set ($n=23$: 12 controls, 11 patients): Top-6 montage (top row) vs. Top-18 (bottom row). Errors are dominated by false positives (top-right cell); false negatives (bottom-left) are rare. Chronos@18 reaches zero false negatives; the Top-18 montage chiefly increases true negatives (fewer false alarms). Sens/Spec annotated per panel.

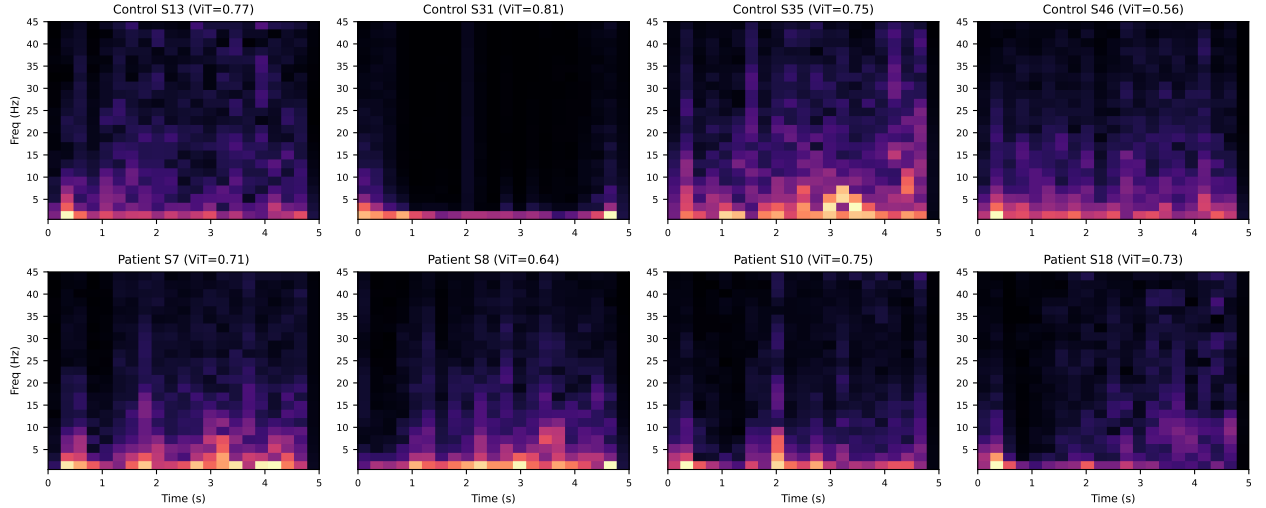
Qualitative STFT spectrograms (Top-6 mean) — controls vs patients, with mean predicted $P(\text{patient})$ 

Figure 7. Qualitative STFT spectrograms (Top-6 channel mean) for four control (top row) and four patient (bottom row) test subjects, annotated with the ViT’s mean predicted $P(\text{patient})$. Patient spectrograms tend to show more diffuse, higher-entropy energy; ViT scores follow the visual cues, and errors coincide with the visually ambiguous cases.

augmented Dickey–Fuller statistic (F7), which formally tests for stationarity. The latter is particularly telling: EEG is non-stationary by nature, and the appearance of a stationarity test among the most discriminative features indicates that patients and controls differ in *how* their signal departs from stationarity—a property that no single moment captures and that connects directly to the ARIMA/differencing material of the course. The convergence between purely statistical features and known

pathophysiology endows the system with the interpretability that is indispensable for clinical translation.

D. Time–frequency resolution and the wavelet result

That the CWT did not beat the STFT deserves a principled reading. Any time–frequency representation is bound by the Heisenberg–Gabor uncertainty principle: one cannot localize a component arbitrarily well in both time and frequency. The

Training vs. validation loss (18-ch STFT run); dotted line = best-val epoch (early stopping)

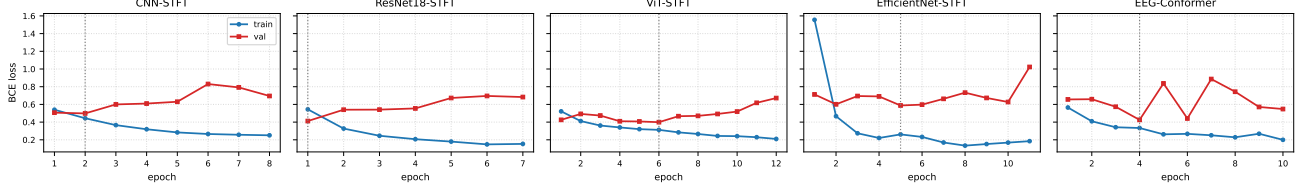


Figure 8. Training vs. validation BCE loss for the deep models (18-channel STFT run). The dotted line marks the best-validation epoch (early stopping). The custom CNN and EfficientNet overfit early; the pre-trained ViT/ResNet18 remain stable for longer, the optimization-level signature of successful transfer.

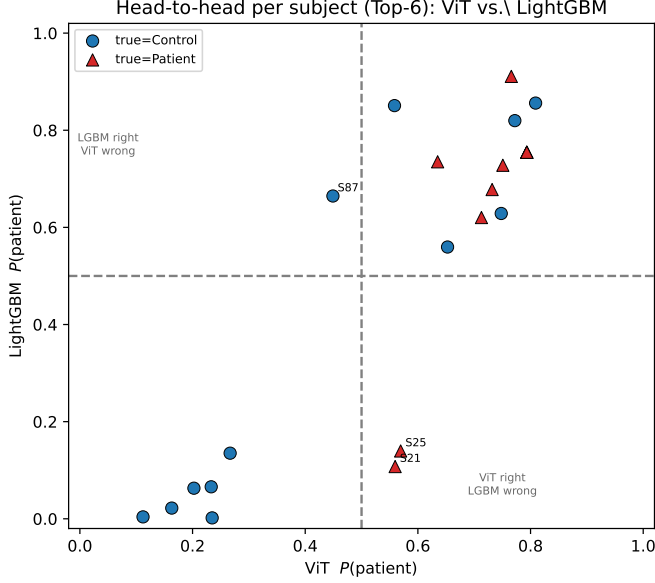


Figure 9. Per-subject head-to-head (Top-6): EEG-Conformer vs. SVM predicted $P(\text{patient})$. Off-diagonal quadrants (labeled subjects) are cases where one model is right and the other wrong; each wins two, revealing a complementary error structure that an ensemble could exploit.

STFT fixes this trade-off with a constant window, whereas the CWT adapts it (fine frequency resolution at low frequencies, fine time resolution at high frequencies). The wavelet advantage is greatest for transient, broadband phenomena; for the relatively stationary, band-limited (0.5–45 Hz) rhythms analyzed here over 5-second windows, the fixed STFT resolution is already adequate, and the additional flexibility of the CWT does not translate into discriminative gains—while its computational cost is far higher. The unexpectedly strong FFT result at the subject level reinforces this: when predictions are pooled per subject, the stationary spectral envelope (which the FFT captures globally) carries most of the signal, and the temporal detail of the STFT/CWT becomes secondary.

E. Window-level vs. subject-level evaluation

Every model improves under subject-level aggregation, and the gain is not cosmetic: averaging the per-window probabilities of a subject reduces the variance of the estimate, behaving as an ensemble over correlated samples. This is also the only clinically defensible reporting unit—a diagnosis concerns a patient, not a five-second epoch—and we therefore advocate

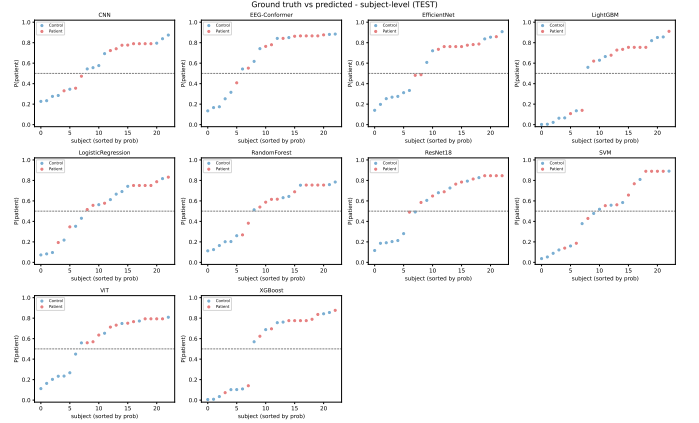


Figure 10. Ground truth vs. predicted probability, aggregated per subject (Top-6). Subjects are sorted by probability; color encodes the true class and the dashed line marks the 0.5 decision threshold. Good separation groups controls (blue) low and patients (red) high.

subject-level metrics as the primary outcome in EEG studies, with window-level metrics reported only as a complementary, higher-variance view (Fig. 10).

F. Limitations and threats to validity

Three limitations temper our conclusions. First, the test set contains 23 subjects; although the subject-wise protocol makes the estimate honest, its confidence intervals are necessarily wide, and external validation on independent cohorts is required. Second, the CWT ablation is confounded by channel count (six vs. eighteen); a channel-matched re-run would settle the wavelet comparison definitively. Third, all results derive from a single dataset and a single resting paradigm; generalization across acquisition hardware and paradigms remains to be established. None of these threats affects the methodological core—the leakage-free, subject-wise protocol—but they delimit the strength of the clinical claims.

VII. CONCLUSION

We presented a rigorous, leakage-free comparative study for EEG-based schizophrenia classification on the ASZED dataset. Methodologically, we showed that a strict subject-wise partition and subject-level aggregation are indispensable for an honest evaluation. Clinically, the EDA quantified cortical disconnection and hyperfrontality, and the SHAP analysis corroborated them independently. On the modeling side, the central result is

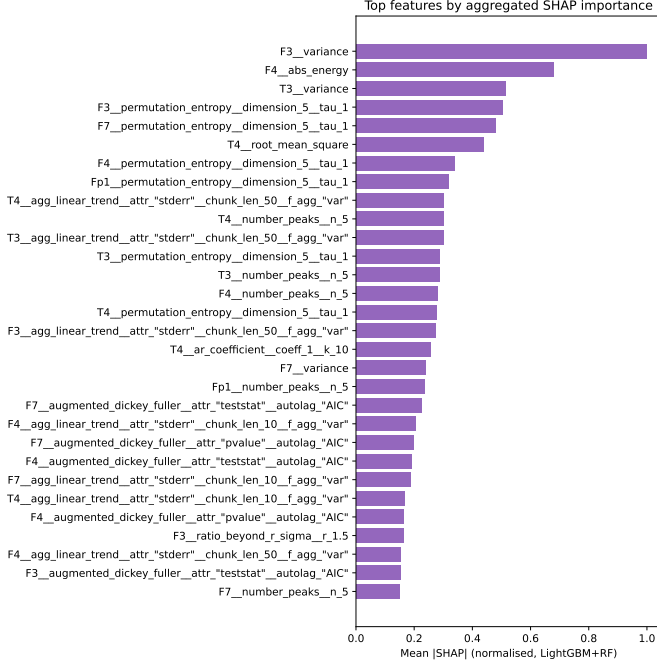


Figure 11. Feature importance by aggregated SHAP value (Top-6). The dominant features belong to fronto-temporal channels, consistent with the hyperfrontality observed in the EDA.

that the *Chronos* foundation model, used frozen and zero-/few-shot, delivers the best overall performance ($F1 = 0.846$, Accuracy = 0.826, ROC-AUC = 0.826), surpassing bespoke deep networks; that the discriminative information is sufficiently fronto-temporal for six channels to suffice; and that wavelet representations do not outperform the STFT/FFT. Interpretable classical pipelines remain competitive, auditable baselines. Future work includes fine-tuning *Chronos* on EEG, channel-matched wavelet comparisons, multimodal time–frequency fusion, and external validation on independent cohorts.

APPENDIX: REPRODUCIBLE IMPLEMENTATION

The complete implementation—eight Python modules (`1_preprocessing.py` through `8_evaluation.py`), centralized configuration, local and SLURM-cluster orchestration scripts, and the theoretical supplement—is released as open source at:

<https://github.com/ricardoamiel/EEG-signal-prediction-for-schizophrenia-detection>

The repository ships the `.parquet` feature cache, the per-configuration artifacts (`top6/top18`), and fixed random seeds to reproduce the reported tables exactly.

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